

MARCEL KÜCK

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RESEARCH INTERESTS

Physical intelligence and embodied AI, with a particular focus on tendon-driven and musculoskeletal actuation. Interested in the intersection of engineered biological tissue and robotic control: how learned controllers can be coupled to soft, biologically inspired actuators. Prior work includes neural-network-based control of a tendon-driven humanoid arm, imitation learning for bimanual manipulation, and hardware–software integration for humanoid platforms.

EDUCATION

Technical University of Munich, *Munich, Germany* 10.2019 – 05.2024
B.Sc. Computer Science. Specialization in Robotics and Artificial Intelligence.

- Bachelor's thesis: *Implementation and Analysis of a Neural Network-Based Control Algorithm for Optimizing Joint-Space Trajectory Tracking for a Musculoskeletal Tendon-Driven Humanoid Robot Arm*. Conducted at Devanthro; supervised by the Chair of Robotics, Artificial Intelligence and Real-Time Systems (Prof. Alois Knoll).
- Member of TUM.AI, Software Development department.

University of Oxford, *Oxford, United Kingdom* 05.2024 – 07.2024
Machine Learning Summer School. Coursework and research seminars on reinforcement learning, generative AI, and machine learning in medicine. Selective admission (7%).

Google Machine Learning Bootcamp Europe 07.2022 – 11.2022
Three-month programme on applied machine learning. Selective admission (~10%).

RESEARCH EXPERIENCE

Devanthro, *Munich, Germany* 03.2026 – Present
Affiliated Researcher (project-based)

- Conducting independent research on control and embodied AI for Robody 5.0, the next-generation humanoid platform built on the OpenArm bimanual robotic arm.
- Building an end-to-end pipeline from VR-based teleoperation through LeRobot policy training to on-hardware deployment.
- Investigating data-efficient imitation learning for bimanual manipulation.

Devanthro, *Munich, Germany* 10.2022 – 08.2024
Research Student & Founder's Associate

- Conducted research on neural-network-based control for tendon-driven humanoid arms (Robody 3.0), culminating in the bachelor's thesis on joint-space trajectory tracking for a musculoskeletal tendon-driven arm.
- Co-developed Robody 4.0: assisted in mechanical assembly, contributed CAD work in Fusion 360, and authored a Python script for parametric design of the elastic hull ("skin") of the Viper arm.
- Supported the founders across fundraising, strategy, and general operations as Founder's Associate.

INDUSTRY EXPERIENCE

MDK Engineering, *Munich, Germany* 01.2026 – Present
Founder

- Solo AI and robotics consultancy providing AI integration and workflow automation as a service to SMB and enterprise clients across multiple sectors.
- Designs and delivers retrieval-augmented generation systems, conversational agents, and AI-powered internal tools tailored to client domains.
- Builds automation pipelines on n8n and Make to integrate LLM and ML capabilities into client business processes and operations.

Rechnungs-API (rechnungs-api.de), *Munich, Germany*
Co-Founder & CEO

02.2026 – Present

- Co-founded a B2B invoicing API; built and operated with a two-person team.
- Active paying customers; approximately 1,000 daily visits.
- Responsible for product direction, technical architecture, and go-to-market.

ShareYourSpace (shareyourspace.com), *Munich, Germany*
Co-Founder & COO

02.2025 – 01.2026

- Co-founded a marketplace for shared and flexible office space, an Airbnb-style platform for the coworking sector.
- Assumed day-to-day operational and product leadership during the company's final year.
- Led the sale of the company and software platform to Pixida GmbH, an existing client.

Siemens AG, *Munich, Germany*
Software Engineer, Siemens SGI

09.2021 – 12.2023

- Designed and shipped a recommender system serving 50,000+ digital resources to 300,000+ users.
- Built machine-learning and web features across the SGI platform.
- Stack: Python, NumPy, TensorFlow, SQL, TypeScript, JavaScript, Angular.

PERSONAL PROJECTS

RustyRobots: Open-source robotics framework for embodied AI in Rust; modular crates for perception, control, and simulation integration. [\[link\]](#)

RustML: Open-source machine-learning framework in Rust; tensor operations, automatic differentiation, and inference. [\[link\]](#)

LeRobot SO-101 Manipulator: End-to-end vision-language-action policy training and on-hardware deployment on the SO-101 arm using the LeRobot framework. [\[link\]](#)

Roasterio: Smart coffee-bean roaster combining a custom sensor stack with automatic roast-profile optimization.

LaMarcello: Retrofit controller converting classic espresso machines into smart, app-connected appliances; PID temperature control and flow profiling.

Murph: Online medical consultation platform pairing medical students with elderly users for accessible primary advice.

Marie Lou Coffee: Personal coffee-roasting venture; small-batch artisanal roasting, exploring a commercial launch.

TECHNICAL SKILLS

Programming: Python, Rust, C++, TypeScript, JavaScript

Machine Learning & Robotics: PyTorch, TensorFlow, LeRobot, ROS 2, NVIDIA Isaac Sim, NVIDIA Cosmos

Hardware & Tooling: Linux, Git, Docker, Fusion 360, SLA printing, Jetson, sensor integration

LANGUAGES

German (native) • English (fluent, C1–C2)